

AFTERSHOCK

Shoimya Chowdhury, Madhura Gujar, Yash Thakare and Yashika Bhoge

October 2026

Summary

Aftershock v1 forecasts how much the S&P 500 will swing over the next five trading days. It does not predict whether prices go up or down, only how bumpy the ride will be. The project builds on *volatility clustering*: turbulent weeks tend to follow turbulent weeks, and calm weeks tend to follow calm ones.

Goal

Answer one question: **does machine learning beat a simple HAR model at forecasting volatility?** If it does not, that is still a useful and honest result.

Approach

Everything runs in a single Jupyter notebook (laptop or Google Colab) using only free tools.

1. **Data:** Daily SPY prices from Yahoo Finance via `yfinance`, from 1993 onward.
2. **Features:** Daily, weekly and monthly volatility (annualized %), plus the previous day's return.
3. **Models:** Naive (next week looks like last week), HAR (a linear blend of daily, weekly and monthly volatility), and LightGBM.

4. **Testing:** Train on 1993–2004 and test on 2005 onward, covering 2008, 2020 and 2022. Retrain yearly on all past data, with strict safeguards against look-ahead leakage.
5. **Evaluation:** Average error in percentage points, plus a chart of forecasts against actual volatility.

Value

- **Learning:** Covers core financial modeling skills, especially leak-free backtesting.
- **Risk:** Volatility forecasts guide position sizing, since holding less when turbulence is expected limits losses.
- **Extensibility:** Later versions could add more tickers, other forecast horizons, better scoring methods, or a simple volatility-aware strategy.

Scope

Aftershock v1 is a learning project, not a trading system or investment advice. Backtests typically look better than real-world results.

[3] [2].

And in the last paragraph write about the proposed deliverable aimed from this thesis [4].

Proposed Guides: Dr./Mr./Ms Name of faculty, Professor/Associate/Assistant Professor, Department, Campus.

References

- [1] N.J. Higham. *Handbook of writing for the mathematical sciences*. Society for Industrial Mathematics, 1998.
- [2] J. M. Linhart. Algorithm 885: Computing the logarithm of the normal distribution. *ACM Transactions on Mathematical Software*, 35:Article 20, 2008.

- [3] Jean Marie Linhart. Teaching writing and communication in a mathematical modeling course. *PRIMUS*, 24(7):594–607, 2014.
- [4] Wikipedia. Fractal, 2012.